



Institute of Engineering and Technology (IET)

Minor Project Mid-Term Report

Project Title

Secure AI Consultation Platform

PREPARED BY

Sarvesh Gulgulia (2023BTech073)

Gaurav Singh Bora (2023BTech032)

Vansh Bhardwaj(2023Btech094)

FACULTY GUIDE

Dr. Raushan Kumar

Assistant Professor | Computer Science and Engineering

March 2026

TABLE OF CONTENTS

1. **Abstract**
2. **Introduction**
3. **Problem Statement**
4. **Methodology and Theoretical Background**
5. **Architecture and Key Design Decisions**
6. **Work Completed Till Mid-Term**
 - 6.1 Research and Design
 - 6.2 Backend
 - 6.3 Frontend
7. **Plan For Remaining Project Timeline**
8. **Risks and Mitigation Strategies**
9. **Appendix**
 - A.1:
 - A.2: References

1. Abstract

Artificial intelligence has significantly improved the ability of computer systems to interact with humans using natural language. Chatbots powered by artificial intelligence are increasingly being used in several domains, including healthcare, to assist users by providing quick responses to their queries.

The Secure AI Consultation Platform is designed to provide preliminary healthcare guidance through a conversational chatbot interface. The system enables users to ask health-related questions and receive automated responses generated by an AI chatbot system. The platform includes a landing page, authentication system, chat interface, and conversation history functionality that allows users to view previous chat sessions.

At the current stage of development, the frontend and backend components of the platform have been successfully implemented. The system allows users to log in, start new chat sessions, and interact with the chatbot interface. The integration and training of the AI model will be completed in the later phases of the project.

The system architecture is designed to be scalable and secure, with future deployment planned using modern cloud technologies such as Docker containers and Kubernetes orchestration.

2. Introduction

Healthcare systems often face several challenges such as limited access to medical professionals, long waiting times, and difficulty in obtaining immediate medical advice. Many individuals seek quick guidance for minor symptoms but may not always have access to reliable healthcare information.

With advancements in artificial intelligence, conversational systems such as chatbots have emerged as a promising solution for providing automated assistance. These chatbots can process user input in natural language and provide appropriate responses based on predefined logic or trained AI models. The Secure AI Consultation Platform aims to provide an accessible and user-friendly system where users can interact with an AI chatbot to receive preliminary health guidance. The system allows users to describe symptoms and ask medical questions through a chat interface. The chatbot then processes the input and provides relevant responses.

If the system detects that a query may require professional medical attention, the chatbot can recommend that the user consult a healthcare professional. The project focuses on developing a scalable and secure chatbot platform that provides an intuitive user experience while maintaining conversation history for users.

3. Problem Statement

Access to reliable medical advice can be difficult for individuals who require quick information regarding minor health issues. Many people rely on internet searches or unreliable sources for medical guidance, which may lead to incorrect or misleading information. Healthcare systems also face limitations in providing immediate consultation for every minor symptom. As a result, there is a need for an intelligent system that can provide basic guidance to users before they seek professional medical help.

A chatbot-based healthcare platform can help address this problem by allowing users to interact with an automated system capable of answering health-related questions. Such a system should be able to maintain conversation history, manage user sessions, and provide a simple interface for communication. The goal of this project is to develop a chatbot platform that allows users to interact with an AI-based system for preliminary healthcare guidance.

4. Methodology and Theoretical Background

The development of the Secure AI Consultation Platform follows a structured and modular development methodology.

- **Requirement Analysis**

The first stage of the project involved identifying system requirements and defining the functionality of the chatbot platform. Key requirements included user authentication, chat interface design, conversation storage, and backend communication.

- **System Design**

After identifying the requirements, the system architecture and database structure were designed. The database entities include **Patient**, **ChatSession**, **ConversationHistory**, **Message**, and **AIChatbot**, which together manage the interaction between the user and the chatbot system.

- **Frontend Development**

The frontend interface was developed to provide an intuitive user experience. The landing page introduces the platform and provides access to the login system. After authentication, users are redirected to the chatbot interface.

- **Backend Development**

The backend server manages communication between the frontend interface and the system database. It handles user authentication, chat session management, message storage, and retrieval of conversation history.

- **AI Integration (Future Phase)**

In the next stage of the project, an AI model will be integrated to process user queries using natural language processing techniques and generate intelligent responses.

Theoretical Background

- **Conversational AI**

Conversational AI refers to technologies that enable machines to communicate with humans through natural language conversations. Chatbots are widely used in domains such as healthcare, customer support, and education.

- **Natural Language Processing**

Natural Language Processing (NLP) is a branch of artificial intelligence that enables computers to understand and interpret human language. NLP techniques will be used in the future to process user queries and generate meaningful responses.

- **Large Language Models**

Large Language Models are advanced AI systems trained on large datasets to understand and generate human-like text responses. These models will be integrated in the later phases of the project to enhance chatbot capabilities.

- **Client-Server Architecture**

The platform follows a client-server architecture where the frontend interface communicates with the backend server through APIs. The backend processes requests and interacts with the database to store and retrieve information.

5. High Level System Design

The Secure AI Consultation Platform follows a layered architecture consisting of the following components:

Frontend Layer

The frontend layer provides the user interface for interacting with the chatbot system. It includes the landing page, login page, and chat interface.

Backend Layer

The backend layer handles authentication, chat session management, and communication with the database.

Data Storage Layer

The database stores user information, chat sessions, and conversation history.

AI Processing Layer (Future Integration)

In future phases, an AI model will process user queries and generate intelligent responses.

5.2 Tech Stack

The platform is built using modern web development technologies.

Frontend

- React and Tailwind

Backend

Node.js and Express.

Database

MongoDB

Deployment (Future Plan)

Docker

ECS

Cloud infrastructure such as AWS

Development Tools:

Git and GitHub

For Model training:

Major libraries: python, torch(gpu version), torchvision.

Concepts: Lora fine-tuning, QLora fine-tuning, fine-tuning using knowledge-graphs, modern optimizers like Lion, bf16.

Pre-trained models used: BioMistral 7B, llama 3.2 – 3B

Additional Design

Pictorial Representation of work

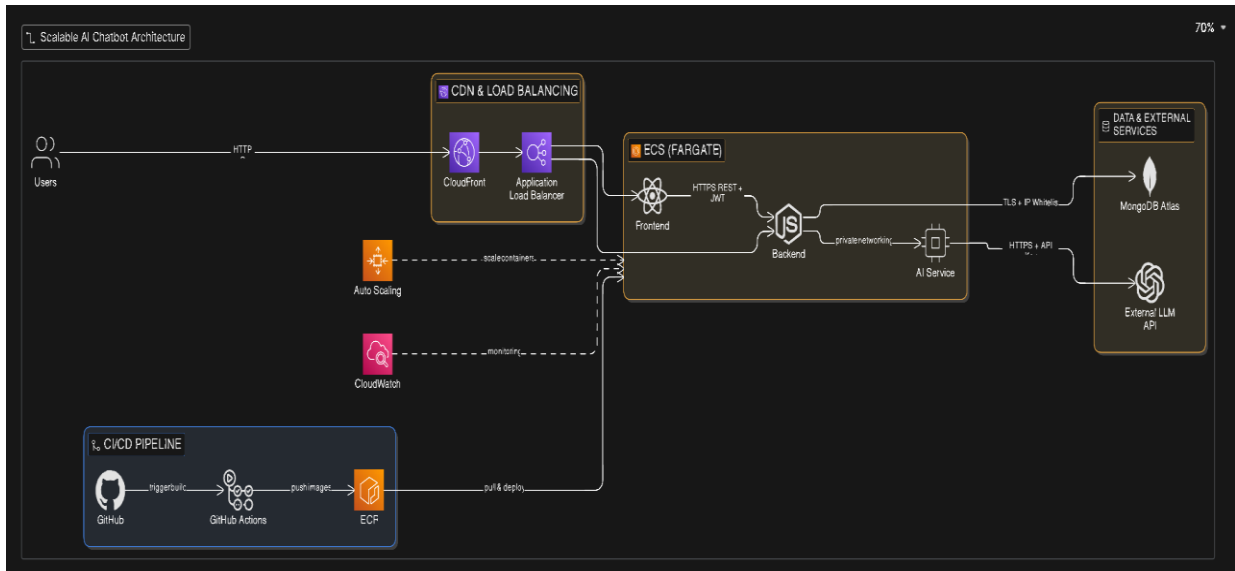


Fig: Architecture of the System

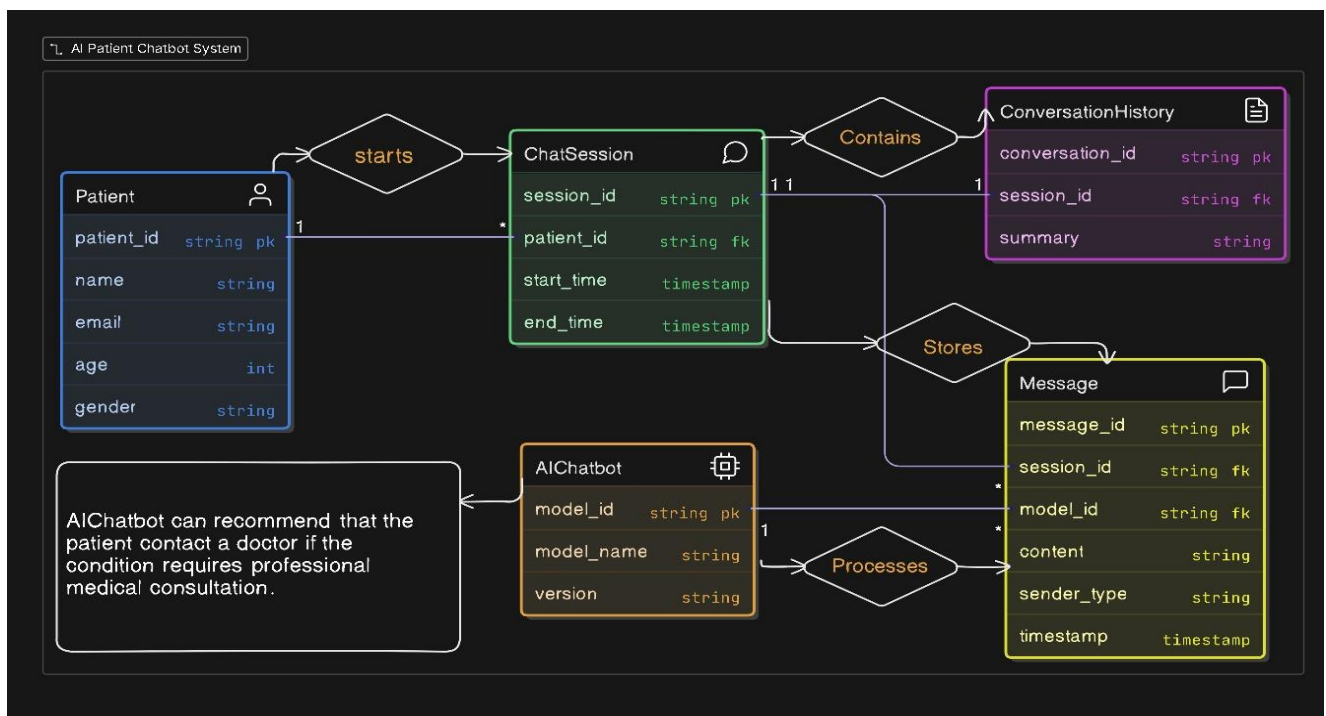


Fig: ER Diagram

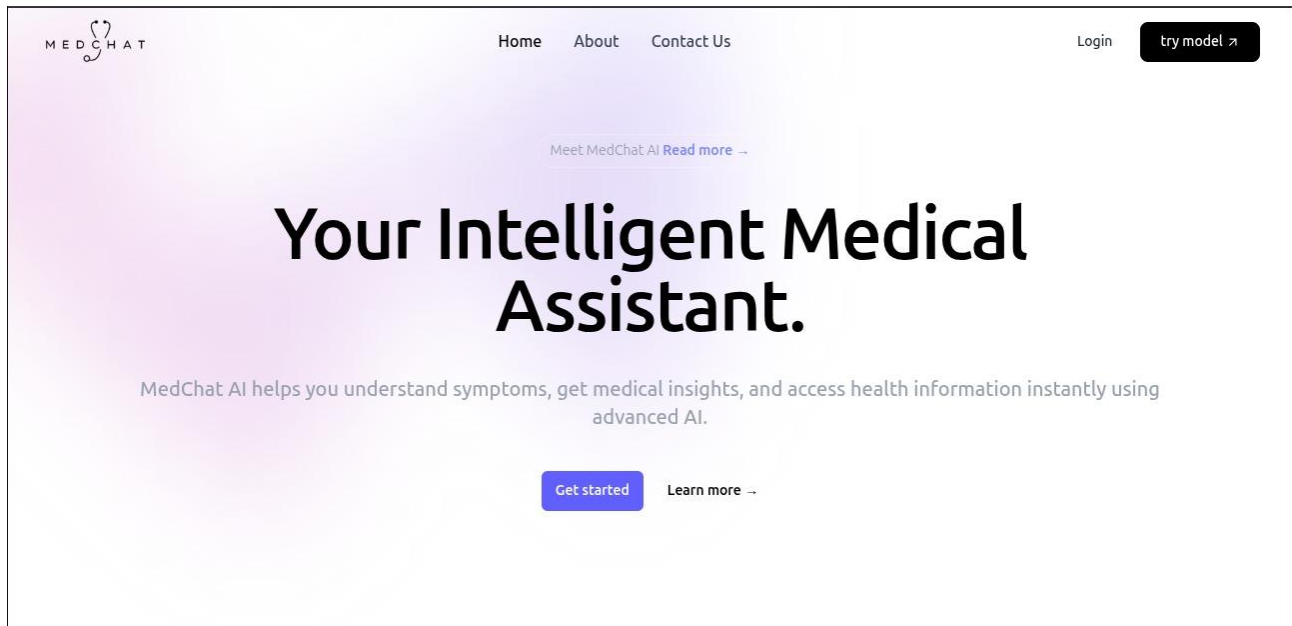


FIG: Landing Page

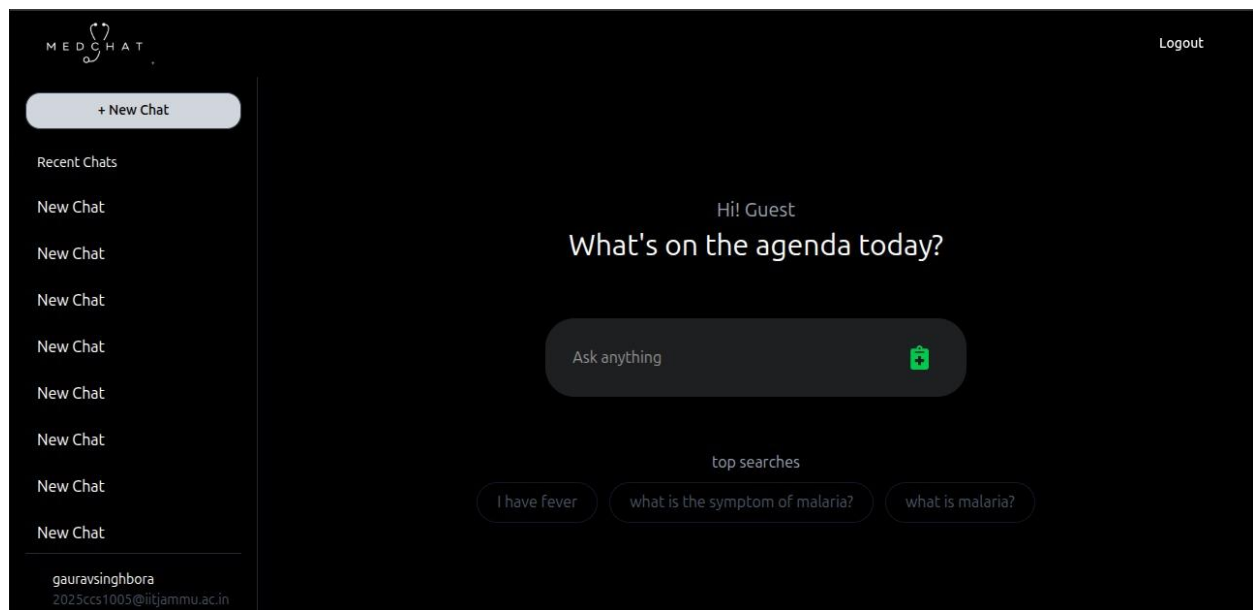


FIG: Home page(After signing in with google account)

6. Work Completed Till Mid-Term

6.1 Research and Design

Initial research was conducted on chatbot systems and conversational AI technologies. Several existing healthcare chatbot platforms were studied to understand their architecture and functionality.

Based on this research, the system architecture and database schema were designed. An ER diagram was created to represent relationships between system entities such as users, chat sessions, and messages.

6.2 Backend

The backend server has been implemented to handle core system functionality.

The backend currently supports:

- User authentication and login system
- Chat session creation
- Message storage
- Conversation history retrieval

The backend architecture is designed in a modular way so that AI model integration can be added easily in the future.

6.3 Frontend

The frontend interface has been developed to provide a user-friendly environment for chatbot interaction.

The implemented features include:

- Landing page introducing the platform
- Login interface for authentication
- Chat interface for messaging
- New chat functionality
- Conversation history panel

The interface is designed to resemble modern messaging platforms to provide a familiar user experience.

Plan for Remaining Project Timeline

The following tasks will be completed during the remaining development period:

1. Integration of AI chatbot model
2. Training or fine-tuning of the AI model
3. Backend integration with AI processing system
4. Cloud deployment using Docker containers
5. Container orchestration using Kubernetes
6. System testing and performance optimization
7. Final documentation and project report preparation

Appendix

A.1 Glossary

AI Chatbot – A conversational software application that interacts with users using natural language.

NLP – Natural Language Processing technology used to process human language.

API – Application Programming Interface used for communication between systems.

Docker – A container platform used to package applications.

Kubernetes – A container orchestration platform used for managing containers.

A.2 References

- I. <https://arxiv.org/abs/1706.03762>
- II. <https://arxiv.org/pdf/2407.21783>
- III. <https://arxiv.org/pdf/2305.10435>
- IV. <https://arxiv.org/pdf/2401.02385>
- V. <https://arxiv.org/pdf/2107.03134>
- VI. <https://arxiv.org/pdf/2402.10373>
- VII. <https://arxiv.org/html/2510.26788v1>
- VIII. <https://moocaholic.medium.com/fp64-fp32-fp16-bfloat16-tf32-and-other-members-of-the-zoo-a1ca7897d407>
- IX. <https://github.com/unslothai/unsloth>
- X. <https://arxiv.org/pdf/2305.14314>
- XI. <https://github.com/karpathy/nanoGPT/blob/master/bench.py>